AR-Mini · Bill of Materials

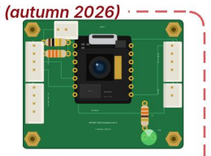
1:9 physical demonstrator of the AmphiReach amphibious rescue robot · what goes into the robot (top) and what builds it (bottom) · status 06.10.2026 (v2)

Compute & Control

Controller v1 (autumn 2026)



Seeed XIAO ESP32-S3 Sense
Wi-Fi/BLE · camera · microSD



XIAO breakout rev 0 (own PCB)
KiCad 9 · AISLER · 50×40



PCA9685 16-ch PWM driver
I²C · ESC + H-bridge PWM



×2

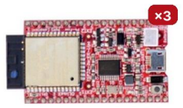
BTS7960 43 A H-bridge
track motors, left / right



×2

Waveshare serial bus servo driver
UART → ST3215 bus

Carrier rev A (next)



×3

Olimex ESP32-S3-DevKit-Lipo
firmware test board



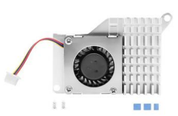
×3

ESP32-S3-WROOM-1 N16R8
16 MB flash · 8 MB PSRAM

Controller v2 · ROS 2 (2027)



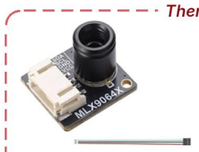
Raspberry Pi 5 · 8 GB
ROS 2 Jazzy · USB to XIAO



Active cooler for Pi 5
fan + heatsink

Sensing

Thermal



MLX90642-D45 thermal camera
forward search · I²C-0



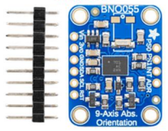
MLX90640 110° thermal camera
32×24 px · cradle · I²C-1

Vision



OV3660 2 MP camera (XIAO Sense)
pan mount on the arch

Inertial

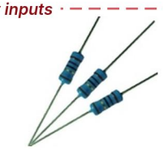


BNO055 9-DOF IMU
heading / roll · I²C-0

Safety inputs



Leak probe — two bare wires
ADC ladder on A1



Battery monitor 100 k / 33 k
VBAT ÷ 4 → A0

Propulsion & Actuation

Water



×2

Cuifati 560 KV deep-water thruster
bidirectional ESC · 12 V

Land



DFRobot ROB0154 tracked chassis
2 DC motors · 152×135 plate

Rescue arm (parallelogram lift)



×4

Waveshare ST3215 30 kg bus servo
magnetic encoder · 12 V bus



×2

TD-8120MG 20 kg servo
fallback option

Camera pan & servo cabling



×8

MG90S 9 g micro servo
camera pan · test rigs



×2

Waveshare 5264 servo cables
ST3215 daisy-chain

Power

Energy



×2

Turnigy 2200 mAh 3S 40 C LiPo
XT60 · 11.1 V



SkyRC iMAX B6 V2 charger
balance charge / storage

Protection & switching



50 A toggle switch (main)
R13-401A2A · OFF-ON



20 A rocker ON-ON (logic kill)
→ PCA9685 OE + ADC



Inline fuse holder GT4652
30 A main fuse



6-slot blade fuse block
20/15/10/5/5 A branches



Blade fuse kit · 250 pcs
spares

Conversion



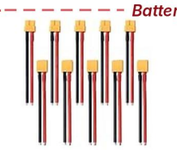
MZHOU 12 → 5 V USB-C buck
logic rail (XIAO, sensors)



QIQIAZI 10–90 V → 5 V 5 A buck
Raspberry Pi 5 rail

Wiring & Connectors

Battery side



XT60 pigtails · 10 pcs
12 AWG silicone



×4

14 AWG silicone wire
black + red · 2 m each



Insulated spade connectors
switches & fuse block

Logic side



×4

18 AWG silicone wire
black + red · 2 m each



JST-XH 2.54 kit + crimped leads
2–6 pin · 22 AWG



Alkan wire ferrules · 800 pcs
0.5–10 mm² terminals



Heat-shrink tubing box
Ø1.5–13 mm

Structure & Materials

3D printing (Creality K2 Pro)



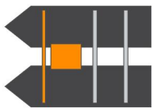
×7

Creality Hyper PLA 1 kg
6 colours, colour-coded



×4

Creality Hyper PETG 1 kg
hulls & all wet parts



Printed hulls, beams & brackets
own CAD, OpenSCAD + F360



Rescue dummy "Pesho" (red PLA)
thermal target

Frame & enclosure



×2

Kantoflex Al profile 15 × 15 × 1
cross-beams · 304 mm



Aluminium rod Ø5 mm
cradle 220 × 123 mm



LANUCN IP65 box 150 × 110 × 70
logic enclosure



×10

WEDZE hand warmers
dummy heat source

Fasteners & bonding



ruthex threaded inserts M2–M5
+ 100 × M3S short



DSM DIN 912 screw set · 640 pcs
M3–M6 · + M2 set · nyloc

Tools & Workshop

Fabrication



Creality K2 Pro Combo
300³ mm build



Digital caliper
0–150 mm

Soldering & crimping



FNIRSI HS-03 iron
2600 mAh



Third hand ZD-126-4
LED + iron stand



Toolour third hand
magnetic base



YIHUA 948DQ-I
fume fan + filters



Solder Sn60Pb40
0.8 mm · 100 g



×3

TOPNIK-ZEL flux
RMA gel



KNIPEX NexStrip
strip · cut · crimp



Preciva crimp set
Dupont · JST-XH

Measurement & debug



UNI-T UT61E+
True RMS



Logic analyzer
24 MHz · 8 ch



Jesverty lab PSU
60 V / 5 A



FT232RL USB-TTL
servo config

Prototyping



×2

VooGenzek PCB kit
perfboards +



Hailege jumper kit
2 × 140 U-wires



Alligator clips
10 pcs



Cleqee clip leads
clip → Dupont

Documentation



NEEWER magic arm
table clamp



ULANZI MT-79
tripod 204 cm



ULANZI VL49
LED light

Software

Design

Fusion 360 · OpenSCAD · KiCad 9 · Creality Print

Firmware & software

PlatformIO · Arduino IDE · Python · Git + GitHub · Pi OS + ROS 2 · PulseView

Services & media

AISLER · Eurocircuits · Printables · SlideRoom · DaVinci Resolve

Sources: AR_BOM_06_10_2026_v3.xlsx · product photos and logos from the shop listings / vendors (Amazon.de, Comet.bg, Elimex, Vikiwat, NenovTechnic, Ardboard) · items without a photo are drawn schematically. ×N = quantity ordered.
Notes: v1 = XIAO-only controller (autumn 2026) · v2 = Raspberry Pi 5 + ROS 2 (2027). Printed parts: PETG for everything wet, PLA for dry parts (orange = moving, black = structure, red = dummy, blue = enclosures). AmphiReach · AR-Mini · T. Kyoseva, Sofia, 2026